Zero empty ATMs, shorter routes

Forecast-driven cash-in-transit routing for Siam Northeast Bank

–8% CIT cost per day · cash-out risk 64 → 0 nodes · THB 1.5M a year

Khon Kaen, Northeast Thailand · 159 branches · 520 machines · 3 cash centres · 30 vans

Siam Northeast Bank (SNB) is a fictional bank. All data is synthetic; every figure in this deck is a synthetic-data demo result, not a customer result.

The outcome, in the buyer's KPIs

Indicative annual CIT saving: THB 1.5M, before idle-cash carry cost or avoided cash-outs. Fixed 5-day cycle vs demand-driven plan, same estate, **1 Oct 2026** (synthetic plan day).

Buyer KPI	Fixed 5-day cycle	Forecast + optimizer	Change
CIT cost / day (THB)	75,400	69,550	**–8%**
Vans dispatched	11	13	+2 crews to hit SLA
Route km			**–23%**
Cash-out risk nodes	64	0	visited in time
Idle cash released (THB)		43M	float freed
Time-window SLA		100%	inside opening hours
Forecast WAPE	naive 39%	model 25%	better signal

The problem: a calendar, not demand, decides where cash goes

SNB replenishes every branch, ATM and cash-recycling machine (RCM) on a **fixed 5-day cycle**.

Symptom	What it costs SNB
Wasted stops: vans visit machines that are still mostly full, because "it's day 5"	Fuel, armed-crew hours, vehicle wear
Cash-outs: payday and holiday spikes empty ATMs before the next visit	Lost fees, branch queues, complaints, brand damage in rural districts
Idle cash: cassettes topped up too early, vaults overstocked "just in case"	Non-earning float, insurance, vault capacity pressure
Static routes: the same sequence every cycle	Excess km, overtime, missed bank-opening windows

A spreadsheet and radio dispatch cannot jointly answer *who needs cash tomorrow* and *which vans can cover them before 18:00*.

Who owns the decision, and what each gets

Executive sponsor: CFO / COO

- Lower **cost-to-serve** per machine: fewer vans and km for the same network
- Less **idle float**: cash that can be redeployed rather than funded
- **Fewer cash-outs**: protected fee income and customer trust
- One governed platform instead of spreadsheets plus point tools

Domain owner: Head of Cash Operations / CIT dispatch lead

- Tomorrow's plan ready every morning: who needs cash, how much, which van, what order
- A map of every route against the old cycle; **override** any stop with a reason, audited
- Ask in plain language: *"Which ATMs in Ban Phai run out before Friday?"*
- Re-forecast one machine on demand when an event or holiday is announced

How it works: one integrated data journey

```mermaid

flowchart LR

A[Synthetic raw CSVs<br/>UC Volume] -->|Auto Loader| B[Lakeflow SDP<br/>bronze → silver → gold<br/>expectations]

B --> C[Unity Catalog<br/>tags · PII mask · depot row filter<br/>grants · lineage]

C --> D[LightGBM forecast<br/>MLflow · UC registry<br/>Serving snb-cash-forecast]

D --> E[Replenishment orders<br/>P80 safety stock]

E --> F[OR-Tools CVRP-TW<br/>route optimizer job]

F --> G[Lakebase Autoscaling<br/>synced routes + node status<br/>dispatch_overrides]

F --> H[Genie space<br/>SNB Cash Ops Assistant]

G --> I[Databricks App<br/>snb-cash-route]

H --> I

D -->|re-forecast| I

```

Each stage writes the next stage's input. One **Databricks Asset Bundle**; one job `snb_cash_ops_e2e` runs raw → app-ready, all **serverless**.

Lakeflow + Unity Catalog: trusted cash data, governed once

Lakeflow Spark Declarative Pipelines (`snb_cash_ops_pipeline`)

- Auto Loader ingests 8 raw feeds (machines, branches, daily cash, CIT trips …) incrementally
- Silver **expectations** drop bad keys and negative cash; warn on coordinates outside Thailand
- Gold: unified node cash flow, demand features, node status, baseline KPIs

Unity Catalog

- Custodian name and phone **masked** (PII) for anyone outside `snb_pii_readers`
- **Row filter** by depot: a dispatcher in one cash centre sees only that centre's trips
- Layer, domain and PII **tags**; gold-only grants; **lineage** from raw file to app

Why Databricks: quality rules, security and lineage are declared once and enforced for SQL, ML, Genie and the App alike. No copies, no second security model.

Intelligence: forecast demand, then solve the routes

Forecast

- **One global LightGBM** across 679 nodes: calendar, payday, holiday, lag and rolling features
- Point forecast **plus a P80 quantile**: replenish for a busy day, not an average day
- Time-based holdout (Aug–Sep 2026): **WAPE 25% vs 39% naive**
- **MLflow** tracking → **UC** model `cash_demand_forecast@champion` → **Model Serving** `snb-cash-forecast`

Optimize

- Forecast → **replenishment orders** (DELIVER / PICKUP, amount, CRITICAL / HIGH / NORMAL, time window)
- **OR-Tools CVRP-TW** per depot: van cash capacity, opening hours, 07:00–18:00 shift, 25 s solve
- CRITICAL stops carry a very high drop penalty, so they are almost never skipped

Why Databricks: features, model, registry, serving and the optimizer share the same governed tables. The App calls the same endpoint the overnight plan used.

Lakebase, Genie, App: in the dispatcher's hands

Layer	Role
Lakebase Autoscaling `snb-cash-ops`	UC **synced tables** of route stops and node status for millisecond reads; writable `dispatch_overrides` (SKIP / ADD / RESEQUENCE / EMERGENCY, reason, author)
Genie space "SNB Cash Ops Assistant"	Agent mode over governed gold tables: *"which nodes cash out in 3 days?"*, *"cost vs baseline?"*. Masks and row filters still apply
Databricks App `snb-cash-route`	Map of optimized vs baseline routes, KPI cards, order list, **Re-forecast** (serving), override form, embedded Genie chat

A dispatcher's override lands in Postgres with a reason, not in a chat screenshot.

Why Databricks: operational database, analytics and AI share one identity and governance model. There's no separate app database to secure and hand-sync.

Demo flow (7–10 minutes)

1. **Outcome first**: App KPI cards for cost per day, vans, km, cash-out risk and SLA
2. **Map**: toggle BASELINE vs OPTIMIZED for 1 Oct 2026
3. **Re-forecast** a payday ATM: the serving endpoint returns a 7-day P80
4. **Override**: skip a healthy stop with a reason; the row lands in Lakebase
5. **Genie**: *"Which ATMs are at cash-out risk tomorrow, and which van covers them?"*
6. **Under the hood**: pipeline graph and expectations → UC lineage, mask, row filter → MLflow run and endpoint
7. **Close**: THB 1.5M a year and the pilot ask

Decisions and trade-offs

Decision	Why	Trade-off
Declarative pipeline, not notebooks	Expectations, lineage, incremental by default	Less ad-hoc flexibility
One global LightGBM, not 679 models	Shares signal, handles new ATMs, one endpoint	Less per-node tuning
P80 quantile for replenishment	A cash-out costs more than a little extra cash	Some extra float carried
OR-Tools, 25 s per depot	Open source, explainable, fast	Not provably optimal
Lakebase synced tables for the App	Fast reads plus writable overrides	Snapshot freshness, not streaming
Genie agent mode, not a custom agent	Governed, no prompt plumbing, fast to value	Less custom tool control
Serverless everywhere	No cluster ops, scale-to-zero	Cold starts

Full rationale: `docs/decisions.md`.

Business case (synthetic-data demo, indicative)

Lever	Basis	Value
CIT operating cost	(baseline – optimized) cost per day × 365	**THB 1.5M a year**
Idle cash	Float reduction across 679 nodes	THB 43M released, plus carry cost avoided
Cash-out risk	Risk nodes 64 → 0 per plan day	Fee income and complaints protected
Fleet	Vans 11 → 13 per day	Two extra vans so SLA goes 15% → 100%

Cost model: distance cost plus THB 2,500 per van-day; road km = straight-line × 1.3. These are synthetic assumptions, validated on real SNB data in the pilot before any commitment.

Time to value: the stack deploys from one bundle; the pilot is one cash centre in about 6 weeks.

Path to production

Phase	Scope	Exit criteria
Pilot (6 weeks)	Khon Kaen Central cash centre; real ATM switch, vault and CIT feeds under the same silver contract	Forecast beats current planner; dispatchers accept ≥ 80% of planned stops
Scale	All 3 cash centres; nightly plan plus mid-morning re-forecast on payday	Cost per day and cash-out KPIs tracked weekly
Harden	IdP groups for row filters, maker-checker on overrides, CI/CD for the bundle, model monitoring, road-network travel times	Risk and Security sign-off; PDPA review
Extend	Denomination mix, vault planning, CIT vendor SLAs, other regions	Region-by-region rollout

Humans confirm every plan; nothing moves cash autonomously. Core banking, CIT vendors and radio stay in place.

The ask

Pilot one cash centre (Khon Kaen Central) for 30 CIT days.

Success, against the fixed cycle on the same nodes:

- CIT cost or km down **≥ 20%**
- Cash-out incidents **not higher**
- Dispatchers accept **≥ 80%** of recommended stops

Next step: a 2-hour working session with Cash Ops and IT Security to scope the pilot data feeds.

Repo: {{GITHUB_URL}} · Siam Northeast Bank is fictional; all data is synthetic and every result shown comes from a synthetic-data demo.